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Triangles, Major Powers, and Rivalry Duration

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Abstract

Why do some international rivalries last longer than others? Rivalry literature focuses on the dyadic interactions within rivalries, but most rivalries have been connected to other rivalries. This paper examines a rivalry's overlapping immediate strategic environment focusing on the countries directly connected via rivalries or alliances. Specifically, I consider how rivals are connected to third parties as common enemies, common friends, or mixed. I focus on three mechanisms to explain the third parties' influence on the focal rivalry. First, a rivalry is more likely to end when it has common allies *(information flow)* or common enemies *(prioritization)*. Second, a rivalry with mixed third parties is more likely to endure *(stable blocs)*. Analyses of rivalry duration show that the duration of minor-minor rivalries is more likely to be influenced by common allies and mixed third parties, while that of major-minor rivalries is more likely to be affected by common enemies.

Keywords

international rivalries, triangular relationships, network analysis

Wars and armed conflicts have waned (Holsti 2006; Mueller 2006), and the number of international rivalries has decreased (Diehl et al. 2019). Despite the decrease in militarized conflicts and rivalry relationships, 60 out of 301 rivalries are still ongoing, and rival states are actively involved in militarized disputes.[1] For instance, from 1816 to 1992, 49% of wars and 40% of militarized disputes occurred within enduring rivalries

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(Diehl and Goertz 2000), and among 21 ongoing militarized disputes as of 2010, 17 occurred between rivals, such as conflicts in the East China Sea (China-Japan), the South China Sea (China-Philippines and Vietnam), and Kashmir (India-Pakistan).[2] Given that some rival relationships disappeared while others are still existing, why do some rivalries last longer?

As this paper demonstrates, third-party relationships influence rivalry longevity. Even though the concept of rivalry has been developed based on the characteristics of repeated conflicts between two countries, rivalries have been connected to other countries in various ways. For example, the inter-Korean rivalry has deeply been connected to other rivalries, particularly, the US-China rivalry. First, these two rivalries are entangled via alliances. The US has been South Korea's ally, and China has considered North Korea a strategic partner for decades. Second, both North Korea and South Korea have a rival relationship with their rival's major-power ally, the US and China, respectively. The relationships among the four countries can be divided into triangles. If the inter-Korean rivalry is the base of a triangle, it is embedded in two triangles with two third parties, the US and China. As one's ally and the other's rival, the US and China have influenced the longevity of the inter-Korean rivalry.

If a rivalry is embedded in triangular relationships, how do these triangular relationships influence its duration? Do the relationships between the focal rivalry and third parties matter? Beyond the alliance literature's focus on the existence of alliances for a country (i.e. extended deterrence), this paper focuses on third parties connected to two rivals. Depending on the types of relationships, the third party can be a common ally, common enemy, or mixed. Those third parties show the direct, immediate relationships the two rivals share, which indicates the rivals' *overlapping strategic environment*. I argue that a rivalry's strategic environment consisting of third parties can affect the longevity of the focal rivalry.

Focusing on all types of triangles consisting of rivalries and alliances, I examine the impacts of third parties on rivalry duration, addressing three mechanisms: *information flow*, *prioritization*, and *stable blocs*, combined with the major power status of the third parties. As mentioned above, a third party can be a common ally, common rival, or mixed (one's enemy and the other's ally). First, a common ally can be a potential mediator for the focal rivalry to avoid being dragged into the rivals' conflict and focus on the original purpose of the alliance (*information flow*). Second, a common rival can reduce rivalry duration by changing the importance and severity of previous rivalries (*prioritization*). Third, a mixed third party has the benefit of sharing its burden with its ally to fight against a rival in the focal rivalry, which increases the durability of both rivalries and the alliance (*stable blocs*). From the mechanisms, it is expected that rivalries with common allies or common enemies are more likely to terminate, while rivalries with mixed third parties are more likely to endure.

Fourth, major-power third parties have greater effects on rivalry duration than minor-power third parties. Specifically, major-power common allies can provide more benefits to the focal rivals when they decide to end their rivalry, compared to minor-power third parties. Also, major-power common rivals can generate greater threats, so

the focal rivals are more likely to change their priority. On the other hand, major-power mixed third parties, one's ally and the other's rival for the focal rivalry, have more capabilities to support their ally to fight against the rival in the focal rivalry compared to minor-power mixed third parties. I expect that a rivalry is more likely to terminate when it has more major-power common allies or rivals, while it is more likely to endure with more major-power mixed third parties.

This paper makes several contributions to the rivalry and alliance literature by addressing triangular relationships of rivalries and alliances. First, moving beyond a country's direct dyadic (one's rival or ally) and indirect triadic (the rival/ally of one's ally/rival) relationships, this paper explores a rivalry's overlapping strategic environment by focusing on its shared third parties. Some studies have focused on one type of triadic relationships (e.g. complex rivalries and pivotal deterrence), but countries have been embedded in various types of security triangles. This paper examines all types of third-party connections and tests their effects on rivalry duration. Second, the rivalry literature emphasizes the effects of political shocks on rivalries, which can create new rivalries and end previous rivalries simultaneously. Going one step further from previous findings, this paper shows how political shocks affect triangular relationships across time and regions. Third, the empirical results suggest that the major power status of the focal rivals and their third parties also affects rivalry longevity, which implies that power plays a large role in triangular relationships among countries.

Conceptualizing Rivalries

A rivalry is characterized as a pair of states with hostility, competitiveness, and contentious issues (Bennett 1998; Colaresi et al. 2007; Goertz and Diehl 1995, 2000; Mitchell and Thies 2011; Thompson 2001). Rivalries have been studied because a large percentage of wars occurred between pairs of countries that viewed each other as enemies and engaged in militarized conflicts repeatedly (Mitchell and Thyne 2010; Stinnett and Diehl 2001; Thompson 1995). In the 1990s, Goertz and Diehl (1995, 2000) introduced the concept of enduring rivalries and identified them as a dyad that experiences six or more militarized disputes (MIDs) over a period of 20 years. Another approach identifies strategic rivalries based on a mutual perception of threat (Colaresi et al. 2007; Thompson 2001). On the other hand, some scholars highlight the existence of contentious issues (Bennett 1998; Dreyer 2012; Mitchell and Thies 2011). For example, Bennett (1998) points out that two states' disagreement over the resolution of some issues is the key characteristic of a rivalry. This issue-based approach is partially incorporated into the concept of rivalry in Klein et al. (2008) and Goertz et al. (2016).

The new rivalry concept is expanded to the Peace data (Goertz et al. 2016). In the Peace data, interstate relationships are measured on five scales: severe rivalry, lesser rivalry, negative peace, warm peace, and security communities. Among the scales, the first two categories indicate rivalries. The distinction between severe and lesser rivalries is based on the frequency and severity of militarized disputes. The anchor cases of severe rivalries are US-USSR, India-Pakistan, and France-Germany, and those of lesser

rivalries are Bulgaria-Greece, Colombia-Venezuela, and Russia/USSR-Ottoman Empire/Turkey (Goertz et al. 2016). Contrary to enduring rivalries, rivalries with shorter duration are included in the Peace data (Klein et al. 2006). This paper considers both severe and lesser rivalries as rivalries.

Rivalry Duration

The rivalry literature explains rivalry duration at different levels: systemic, dyadic, and state levels. First, political shocks at the systemic and state levels influence rivalry duration (Goertz and Diehl 1995, 2000; Rasler et al. 2013). Since rivalries are in punctuated equilibrium, dramatic changes at the system and state levels can alter the stable, hostile relationships (Diehl and Goertz 2000). Traditionally, world wars, changes in territorial sovereignty, and shifts in power distribution are considered system-level political shocks, and a state's independence and civil war are considered state-level political shocks (Goertz and Diehl 1995, 2000). Focusing on escalation and de-escalation in rivalries, Rasler et al. (2013) also highlight the impacts of political shocks on rival states' decision-making processes. They specify four types of political shocks: shift in threat perception, change in regime orientation/strategies, change in competitive ability, and domestic resource crises. Recently, disasters are also treated as political shocks influencing interstate rivalries (Akcinaroglu et al. 2011; Lee et al. 2022). Those political shocks are considered as a necessary, not sufficient condition for rivalry termination/de-escalation.

Second, at the dyadic level, the existence and settlement of contentious issues influence rivalry termination (Bennett 1998; Dreyer 2012; Rider and Owsiak 2015). Two rivals with ongoing homeland territorial issues tend to stick to their hostile relationship, but once they reach a compromise over the issues or one of them gives up the issues, that leads to the end of the rivalry (Bennett 1998). Relatedly, border settlement leads to rivalry termination (Rider and Owsiak 2015). However, agreements are difficult to reach if territorial disputes involve highly salient issues; thus, rivalries concerning territorial disputes (particularly homeland territory) are more likely to endure (Bennett 1998; Dreyer 2012).

Third, at the state level, a rival country's security level can affect rivalry duration (Colaresi et al. 2007; Kupchan 2010). If a country has multiple rivals, managing the rivals is a burden for the country, given its limited resources (Colaresi et al. 2007). Also, the leader of the country can make a strategic choice of ending or de-escalating some rival relationships to focus on a bigger threat (Akcinaroglu et al. 2014; Kupchan 2010). Not only the number of rivals but also their relative capabilities bring about the leader's concern about security, which can motivate the leader to change the situation by reducing the number of external rivals (Bennett 1998).

One of the factors Bennett (1998) points out is common threats. Despite his mixed findings, this approach implies that some rivals have had common rivals. The US-USSR-China relationship during the Cold War is a triangle where a rivalry has a common rival. On the other hand, in the aforementioned example, the South Korea and

North Korea rivalry has been connected to the US and China with alliances and rivalries, which makes two triangles with two mixed third parties. Beyond Bennett's (1998) approach to common threats, this paper focuses on all types of triangles, indicating a rivalry's *overlapping immediate environment* consisting of third parties connected to both rivals.

Triangular Relationships in International Relations

A rivalry's common rivals indicate that the focal rivalry is part of a rivalry triangle, where the enemy of my enemy is my enemy. The phrase, "the enemy of my enemy is my friend," is well-known from the structural balance theory (Heider 1946, 1958). According to this theory, cognitive units or entities tend to attain balance, "a situation in which the perceived units and the experienced sentiments co-exist without stress" (Heider 1958, 176). Three entities can achieve structural balance if all relations are positive (the friend of my friend is also my friend) or two of three are negative and one is positive (the enemy of my enemy is my friend). If the three entities have one negative and two positive relations (the friend of my enemy is my friend) or three negative relations (the enemy of my enemy is my enemy), they are in imbalanced situations, and thus, some of the relations will change to attain a balanced situation (Heider 1946, 110).

Structural balance theory has been applied to international relations. First, some international relations literature focuses on the balanced relationships among countries (Lee et al. 1994; Maoz et al. 2007). Using dynamic models, Lee et al. (1994) test how the balance is achieved among countries and show that an imbalanced status can change to a balanced state depending on the strength of ties among them. Maoz et al. (2007) empirically show whether indirect relationships influence conflict or cooperation. They find that alliance is more likely to be formed with indirect friends, and militarized conflicts/wars are more likely to occur with indirect enemies. However, their results also reveal imbalanced relationships among countries: alliance is more likely to be formed with indirect enemies (the friend of my enemy), and militarized conflicts/wars are more likely to occur with the enemy of my enemy. In their analyses, imbalanced dyadic relationships are more likely to occur in strategic rivalries (Maoz et al. 2007), implying that rivalry triangles are not rarely observed among countries.

Second, some rivalry scholars focus on third parties, such as common allies, common rivals, or outside joiners. Valeriano and Powers (2016) coin the term complex rivalries to identify rivalries with more than two countries engaged with common issues, alignments, or joiner dynamics. Complex rivalries are triads where two countries have long-term animosity against their common enemy. They find that complex rivalries are more likely to experience wars than dyadic rivalries (Valeriano and Powers 2016). Akcinaroglu and Radziszewski (2017) consider the connectedness of rivalries by focusing on direct and indirect economic and military alliance ties. They find that a core country's accommodation decisions depend on the rivals' centrality in economic and military alliance networks (Akcinaroglu and Radziszewski 2017).

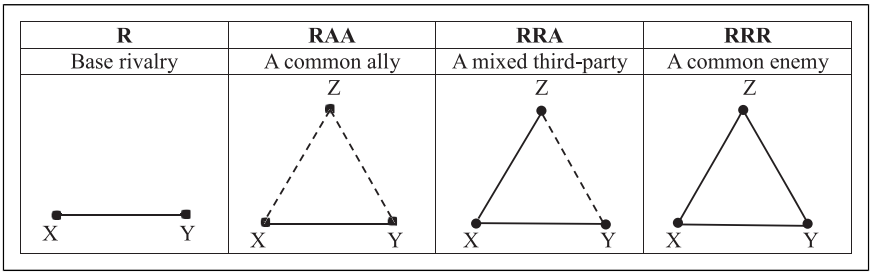


Figure 1. Types of Triangular Relationships. *Note:* A solid line indicates a rivalry (R), and a dashed line denotes an alliance (A).

Among the triangle-related research, complex rivalries are explicitly closed triangles, but others are not necessarily closed triangles (Figure A1). When applying Heider's balance theory to rivalries, a rivalry can have three types of triangles (Figure 1). If the base of a triangle is the focal rivalry (X-Y), the third party (Z) connected to both rivals can be a common ally (RAA, where A = alliance and R = rivalry, e.g. the US for the South Korea-Japan rivalry), common rival (RRR, e.g. Israel for the Egypt-Jordan rivalry), or mixed (RRA, e.g. China for the South Korea-North Korea rivalry). Of course, some rivals do not share any third parties (R, e.g. the India-Sri Lanka rivalry). Valeriano and Powers's (2016) complex rivalries are considered RRA triangles. From previous research, we know that rivalry triangles are common (Maoz et al. 2007), and complex rivalries are more likely to have wars (Valeriano and Powers 2016). However, we still do not know when and where those triangles occurred and how different types of triangles influence rivalry duration.

Using rivalries from the Peace data (Goertz et al. 2016) and defense pacts from the Alliance Treaty Obligations and Provisions (ATOP) data (Leeds et al. 2002), I count the number of each type of triangle from 1900 to 2015.[3] First, the number of RRA triangles surged upward during the First and Second World Wars (Figure 2). Although the spikes are relatively small, the number of RAA and RRR triangles increased during the World Wars. This implies that newly generated rivalries and alliances during wartime created all three types of triangles. In other words, system-level political shocks bring about not only new rivalries (Goertz and Diehl 1995) but also new triangular relationships in international relations.

Second, new triangular relationships were generated during the Cold War, which lasted longer than triangles during the first and second World Wars. The frequency of RRA shows a sharp increase and decrease at the start and end of the World Wars, while it has moderate ups and downs during the Cold War. The increase in RRA during the Cold War can be explained by more newly formed alliances and their longer duration (Leeds and Mattes 2007) and the intensity of the US-Soviet Union rivalry. More precisely, in Figure 2, RRA shows a moderate, steady increase through the mid-1950s and 1960s and a moderate decrease in the 1970s (détente). Again, it increased in the

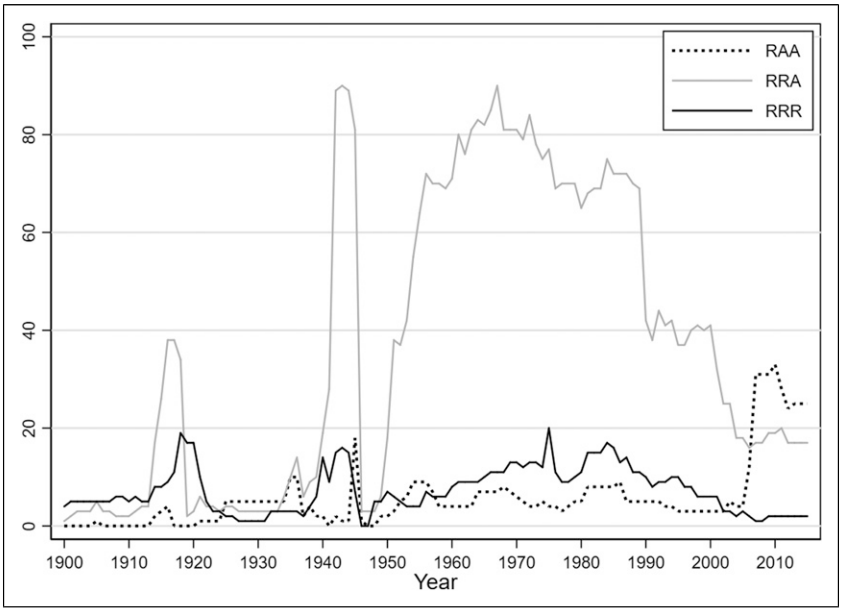


Figure 2. Frequency of triangular relationships (1900-2015).

1980s (the so-called "second Cold War") and then decreased moderately during Gorbachev's rule before the Cold War ended.

Third, when looking at the spatial distribution of triangles between 1900 and 1945 (Table 1), most triangles were generated during the World Wars, so they were composed of some or all European countries. In the aftermath of the Second World War, more triangular relationships emerged with the independence of new countries and decolonization in other regions such as Africa, Middle East, and Asia. For instance, Israel's independence in 1948 brought several rivalry triangles (RRR)—Israel-Egypt-Jordan, Israel-Syria-Jordan, Israel-Saudi Arabia-Jordan, and Israel-Saudi Arabia-Iraq. With the emergence of a new rival (Israel), old rivalries became embedded in rivalry triangles. Those rivalry triangles show that political shocks produce not only new rivalries but also rivalry triangles.

Triangles as Overlapping Immediate Strategic Environment

Some rivalries have more than one third party, which are not necessarily the same type. For instance, the inter-Korean rivalry had mixed third parties, the US, China, and Russia, and a common rival, Japan. All third parties directly connected to both rivals are part of the *overlapping immediate strategic environment* of the rivalry. The term, strategic environment, describes the situation a country is facing and its strategic

Table 1. Frequency of Triangular Relationships by Regions.

		1900-1945	1946-2015	All
Inter-regional	RAA	21	47	68
	RRA	94	123	217
	RRR	29	24	53
Americas	RAA	—	—	—
	RRA	—	1	1
	RRR	3	—	3
Europe	RAA	20	1	21
	RRA	100	30	128[a]
	RRR	30	1	30[b]
Africa	RAA	—	20	20
	RRA	—	7	7
	RRR	—	4	4
Middle East	RAA	—	—	—
	RRA	—	34	34
	RRR	—	8	8
Asia	RAA	—	—	—
	RRA	3	10	13
	RRR	1	14	15
Total		301	324	622

Note: Rivalries are drawn from the Peace data, and defense pacts are drawn from the ATOP data. a, b: Two RRA triangles and one RRR triangle were generated between 1900 and 1945 and lasted for the post-1945 period.

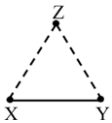
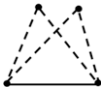
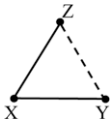
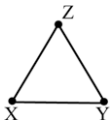
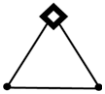
options. Maoz (2011) extends the concept by adding indirectly connected third parties, such as the allies of one's enemies (Maoz 2011). One step further from his approach, this study captures the overlapping area of two rivals' strategic environments. Let's say there are two rivals, X and Y, and Y's allies, Z_1, Z_2, and Z_3. In Maoz (2011), all Zs are considered potential threats to X. This paper focuses on Zs' direct relationships with X. If Z_1 is neither X's ally nor rival, Z_1 is not counted. If Z_2 is X's ally and Z_3 is X's rival, X and Y have one common ally and one mixed third party.

In the next section, I describe how these triangles influence rivalry duration through three theoretical mechanisms: *information flow*, *prioritization*, and *stable blocs*. Additionally, I consider the major power status of third parties in triangles. Compared to non-major powers, major power countries have capabilities and willingness to intervene in other rivalries (Maoz 2011). Thus, I combine the major power status of third parties with the three theoretical mechanisms (Table 2).

Information Flow: Rivalries with Common Allies

In RAA, a rivalry has a common ally. South Korea and Japan were rivals but shared an ally, the US. Although this triangle type is not expected to appear frequently

Table 2. Mechanisms in Triangular Relationships.

Types		Basic Mechanisms		Applications
RAA		Information flows through the common ally (South Korea-Japan-US)		More opportunities for X and Y to get mediated
				Major power Z can provide more benefits to X and Y, when they end their rivalry
RRA		Y and Z cooperate to fight against their common rival, X (South Korea-North Korea-China)		Y has more Zs to fight against X, which makes a bloc
				Major power Z can provide more resources to Y to fight against X
RRR		X and Y prioritize a new threat from Z (Egypt-Jordan-Israel)		More common threats can make the X-Y rivalry less important
				Major power Z can create a bigger threat to X and Y

(Maoz et al. 2007), 17.52% of total triangles from 1900 to 2015 are RAA (Table 1). I argue that common allies have incentives to intervene and mediate their allies fighting each other, and with the help of those common allies, the focal rivalry is more likely to end.

The alliance literature examines deterrence and pacifying effects of direct alliance relationships (Johnson and Leeds 2011; Leeds 2003), but a few studies focus on indirect alliance relationships—the allies of one's allies, or triads (Crawford 2003, Cranmer et al. 2012, and Kinne 2013 are exceptional). First, common allies can provide information to both rivals; thus, information flows between the two. In indirect alliance ties, two countries with no alliance ties but a common ally, can gain information about each other's trustworthiness, reliability, and preferences from their ally (Kinne 2013).

Second, common allies can be a pivot between two rivals, which can bring uncertainty to the rivals about common allies' preferences. In pivotal deterrence, two adversaries with a pivot (common ally) can hesitate to wage war against their enemy and even choose a concessional strategy due to the uncertainty and risk of the common ally's alignment with the enemy (Crawford 2003). Suppose Japan were to escalate a conflict toward South Korea, for example. In that case, it could fear the United States would support South Korea in the conflict, reducing its likelihood of winning. Since the common allies have alliance relationships with both rivals, it is not sure who the ally will support in the conflicts between two rivals. This uncertainty leads the adversaries

not to initiate war, and they can make concessions not to be isolated from the alignment of the pivots and the other side (Crawford 2003).

Third, common allies have the benefits of ending their connected rivalries to focus on the original purpose of their alliances. Due to the obligations in alliances, allies can be dragged into conflicts or wars between their connected rivals. Thus, to avoid being dragged into the conflicts, common allies have motivations to intervene or mediate two disputants (Owsiak and Frazier 2014). The United States has frequently intervened as a mediator when territorial and trade conflicts have escalated between Japan and South Korea. Also, we need to consider why common allies make an alliance treaty with two rivals. When the US became an ally for Japan and South Korea, its treaty was not targeting either of them but considered other threats, such as China and North Korea. Not to be dragged into potential conflicts between two rivals and focus on the purposes of the alliances, the common allies have benefits to terminating the rivalry.

Common allies have incentives and motivations for intervention in the focal rivalry. As Owsiak and Frazier (2014) argue, it is risky for third parties if their allies fight each other. The common allies decide to ally with two rivals to focus on other threats. As addressed in Kinne (2013), the US played a large role when South Korea and Japan signed a bilateral military agreement on the general sharing of military intelligence (GSOMIA). This triadic relationship shows the role of an ally, the US. However, without threats like North Korea and China, it is difficult to explain the US' active conflict management role between the two rivals. In their agreement, all three countries have shared information about North Korea's activities.

From the third party's perspective, having allies who fight against each other can undermine its interests and, thus it is more likely to engage in mediation. Considering that a common ally is a potential mediator, having more common allies means that the focal rivalry has more opportunities to have more than one mediator to help end their historically hostile relationship.

Hypothesis 1: The more common allies a rivalry has, the more likely it is that the rivalry will end.

Prioritizing Rivals: Rivalries with Common Enemies

The alliance literature shows that dyads with common enemies are more likely to be allied (Lai and Reiter 2000; Walt 1987). Contrary to their argument, RRR triangles consist of three rivalries, where two countries with a common rival are also a rivalry. Some studies have examined specific rivalry triangles in international relations, such as the US-Soviet Union-China triangular relationship (Dittmer 1981; Goldstein and Freeman 1990), while a few studies focus on rivalry triangles in a broader context. I argue that the appearance of common rivals changes the focal rivalry's previous strategic environment, and thus, focal rivals have incentives to end their rivalry to focus on common threats.

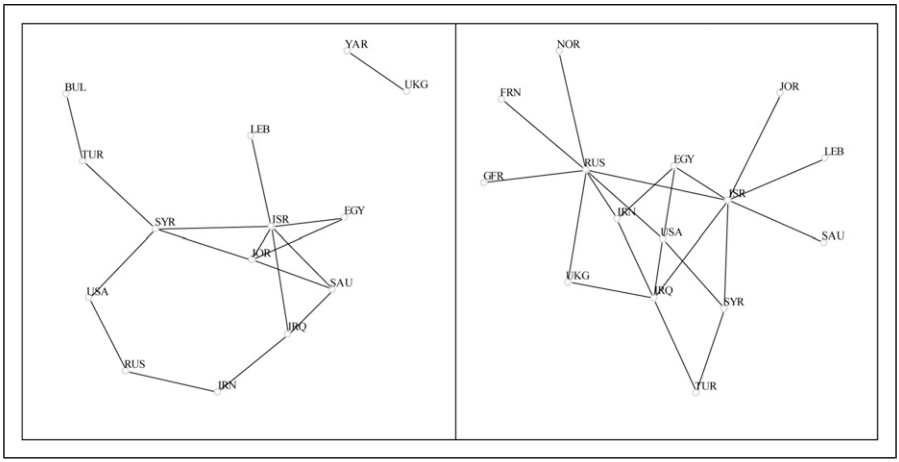


Figure 3. Rivalries in the Middle East in 1950 (left) and 1970 (right). *Note.* Rivalries, including any countries in the Middle East, are presented. Some rivalries which have defense pacts are omitted.

Facing multiple threats can be burdensome for a country. Then, why do those rivalry triangles occur? Most rivalry triangles appeared due to systemic or state-level political shocks (Figure 2). As Goertz and Diehl (1995) find, new rivalries can be formed with political shocks. These new rivalries can be added to old rivalries, which generates rivalry triangles. Suppose X and Y have been rivals. One day, a new neighboring country (Z) appeared, threatening both X and Y. If X and Y were not rivals, they might be allied against a new common threat. However, since X and Y are rivals, both now face two rivals—an old rival and a new rival. Between the two rivals, X and Y need to figure out who is their major threat and determine how they will respond to the threat.

Colaresi et al. (2007) address that decision-makers tend to focus on new threats when facing multiple rivals. They argue that even major power countries have limited resources to deal with multiple rivals. Thus, regardless of their major power status, decision-makers tend to put their old rivals "on the back burner" to focus on their new rivals (Colaresi et al. 2007). In the example above, X and Y might lessen their rivalry as they focus on Z's new threats.

When a rivalry has common rivals, the rivalry is embedded in a rivalry triangle consisting of three rivalries (RRR). Thus, each country in the triangle is two other countries' common rival. In this situation, two countries in the triangle can strategically choose to end their rivalry to focus on their common rival. Figure 3 shows rivalries in the Middle East. The left side (1950) includes four rivalry triangles with Israel. Due to Israel's appearance in 1948, new rivalries between Israel and other countries were formed, which made rivalry triangles. However, on the right side (1970), most triangles disappeared with newly generated alliances in this region. For instance, the Israel-

Egypt-Jordan rivalry triangle was dissolved with the termination of the Egypt-Jordan rivalry in 1967. Egypt and Jordan signed a defense agreement in 1967, considering their common enemy, Israel. Even though Egypt and Jordan had different attitudes toward western countries, the new threat and their loss of territories precipitated a rivalry change (Brand 1994).

Second, the Israel-Syria-Jordan triangle was dissolved. Due to the alliance relationship between Syria and Jordan, their rivalry is omitted in the figure, but the rivalry between Syria and Jordan continued until 2005. The rivalry triangle among those three countries was resolved with a formal peace treaty between Israel and Jordan in 1994, but Ma'oz (1995) points out that Israel and Jordan maintained de facto peace relations since 1970. Even though Israel was a new threat to both Syria and Jordan, the dynamics in the Israel-Syria-Jordan triangle differed from those in the Israel-Egypt-Jordan triangle. For Syria, Jordan was a potential partner in fighting against Israel, and Syria attempted to convince Jordan to join the Egyptian-Syrian military pact in 1956 (Ma'oz 1995). However, Jordan preferred delicate attitudes toward Israel because of its proximity to Israel (Mutawi 1987; Ma'oz 1995). For instance, before Jordan deployed its troops to the Golan front for the 1973 Yom Kippur war, King Hussein contacted Israel to warn about Syria and Egypt's preparations for war (Ma'oz 1995, 155). Israel considered Syria the most dangerous enemy and sought to improve its relations with Egypt and Jordan while maintaining its hostile relations with Syria (Ma'oz 1995, 196). As such, Israel decided to de-escalate its rival relationship with Egypt and Jordan to focus on Syria, consistent with Akcinaroglu et al.'s (2014) finding, and ultimately, those decisions resulted in the rivalry termination.

Rivalry triangles create an environment where every country is all other countries' common enemy. In this environment, rival countries can focus on the most salient and major threats and downplay other rival relationships. If rival countries have more than one common enemy, they could put each other on the back burner and experience more opportunities to terminate their rivalry.

Hypothesis 2: The more common enemies a rivalry has, the more likely it is that the rivalry will end.

Stable Blocs: "Two against One"

One of the significant tendencies in a triad is that two make a coalition against one.[4] When three persons with different levels of power compete over goods, and if they know each other's power, then a coalition against one can be predicted (Caplow 1968). The picture of two against one can be presented as an RRA triangle. In international relations, RRA triangles consist of two rivalries and one alliance dyad. If we twist the triangle, a country (X) has two rivals who are allied (Y-Z). This type is the most frequent between 1900 and 2015 (64.31%, Table 1). I argue that two allies share the burden of fighting against their common rival, which makes their alliance last longer. From the common rival's perspective, it is risky to change the status quo using arms due to an

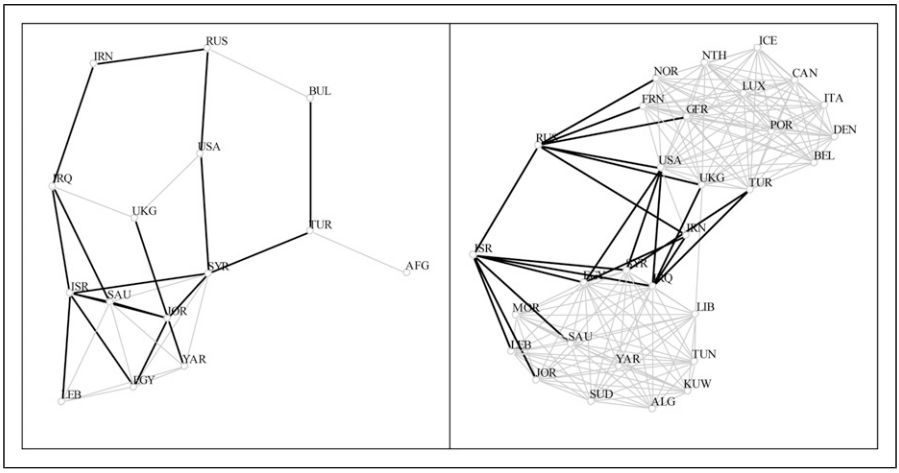


Figure 4. Rivalries and Alliances around the Middle East in 1950 (left) and 1970 (right). *Note.* Rivalries and alliances, including any countries in the Middle East, are presented. Black lines indicate rivalries, and gray lines indicate defense pacts. Some rivalries having defense pacts are omitted.

alliance between their enemies. If the common rival (X) finds its allies outside of the triangle to fight against its rivals (Y-Z), that creates other RRA triangles, which can lead them to maintain their hostile relationships as blocs.

RRA triangles look similar to extended deterrence, which is also two against one (Defender and Protégé against Challenger) (Croco and Teo 2005). From the logic of extended deterrence, it is expected that a challenger (common rival) is less likely to initiate conflicts against a target due to the existence of a defender (Johnson and Leeds 2011). Moreover, some scholars expect the defender's role to restrain the target by advising it to accept a challenger's demand rather than experience the high costs of war (Fang et al. 2014).

In RRA, the existence of an alliance can deter the common rival's attempt to change the status quo, but the defender's role in conflict management might be different in this triangle. In the setting of extended deterrence, Challenger (X) and Defender (Z) are not necessarily rivals, while they are rivals in RRA. For instance, if a defender (Z) is a challenger's rival with issues at stake, then it is difficult for the defender to advise the target (Y) to accept the challenger's demand, which might lead the challenger to change the status quo with the defender as well. Due to its own rivalry relationship with the common rival (X), a mixed third party (Z) in RRA does not have incentives to mediate the focal rivalry and might prefer for them to continue their rivalry to share its burden to fight against its rival. For instance, the US has incentives to mediate the South Korea-Japan rivalry (RAA), but it does not for the Japan-China rivalry (RRA).

On the other hand, the common rival is fighting against two rivals (one alliance) to find its own allies, which brings other RRA triangles. For instance, South Korea has one formal alliance treaty with the US, targeting North Korea. On the other hand, North Korea and China have allied to counter the US-South Korea alliance. Not surprisingly, the inter-Korean rivalry endures because of allied support. When looking at other regions, alliances tend to expand against common enemies. Figure 4 (right) shows that countries in the Middle East allied against Israel. More RRA triangles appeared as cooperative relationships between the bloc members became stronger, and their conflicts against Israel intensified.

In RRA, two allies (Y and Z) share common interests against their common enemy (X). As the two allies' common interests become stronger, their conflicting interests against the common enemy become clearer. Having more RRA triangles indicates a state has many allies fighting against common enemies (NATO against Russia) or has many enemies who are cooperating to fight against it (Russia against NATO). Thus, rivalries embedded in more RRA triangles are expected to endure.

Hypothesis 3: The more mixed third parties a rivalry has, the less likely it is that the rivalry will end.

Power Dynamics within Triangles

When looking at all triangles of rivalry and alliance between 1900 and 2015, 338 out of 622 (54.34%) were inter-regional (Table 1), which means at least one country was located in a different region from the other two. When considering the capabilities and willingness to connect to other regions, major power countries are more likely to connect to rivalries located in another region. Indeed, 300 out of 338 inter-regional triangles (88.76%) include at least one major power country (Table A3).

Major power countries' connection to other regions can occur through alliances or rivalries. In the example of the Israel-Egypt rivalry termination, the US played a crucial role as a major power mediator for Israel and Egypt to decide to end their hostile relationship (Rasler et al. 2013). On the other hand, major power countries can intervene in local or regional hierarchies targeting their rivals, considering their interest (Lemke 2002: 51). Colaresi et al. (2007, 167) also briefly describe the connections between positional and spatial rivalries.[5] Related to the global level competitions (positional rivalries), Colaresi et al. argue that "whales (sea powers) need elephant allies (land powers) to defeat elephant challengers." Their empirical findings support the argument that most major power rivalries rarely have dyadic wars (Colaresi et al. 2007).

Once major power countries are connected to other states' rivalries, rivalry dynamics are hierarchical (Diehl and Goertz 2000). Not all rivalries are at the same level; thus, some are nested (e.g. the US-Cuba rivalry is nested in the US-Soviet rivalry). In the nested structure, major-power rivalries influence one another and lower-level rivalries, while the lower-level rivalries do not influence higher-level rivalries. In other

words, a major-power rivalry can affect changes in its linked major-minor or minor-minor rivalries, not vice versa (Diehl and Goertz 2000).

In RAA, a major-power common ally can guarantee more benefits to the focal rivals that terminate their hostile relationship. The mediation literature points out the importance of major power status for mediation occurrence (Favretto 2009; Greig 2005). Greig (2005) shows that mediation efforts are less likely to occur when one or both disputants are major power countries. He argues that minor-minor rivalries can benefit from major-power third parties, while major-power countries are unlikely to receive mediation from minor-power third parties. He finds that mediation is less likely to be offered or occur for rivalries with at least one major power (Greig 2005).

In RRR, a major-power common rival can change the priority of previous rival relationships more easily. First, the existence of a major-power common rival influences the dynamics of minor power rivalries. Being connected with rivalries indicates that the major power country has issues at stake. Minor-power focal rivals have to pay attention to their major-power common rival. Second, for a major-minor rivalry with a major-power common rival, the two major power countries will tend to focus on each other, thus, one of them will put their rivalry with a minor power country on the back burner. In both minor-minor and major-minor rivalries, a major-power common enemy precipitates the prioritizing process in its rival countries, influencing rivalry termination.

In RRA, a major-power mixed third party has more capability to help its ally in the focal rivalry to fight against their common rival. Particularly, those RRA triangles can be one of the ways an outside major power projects its power to a region's sub-hierarchy (Lemke 2002). As described in Colaresi et al. (2007), to compete with a regional power (elephant), an outside major power (whale) can find an ally in the region, which makes an RRA triangle with a major power. "The star-like nature of China in the field of Asian rivalries" in Ganguly and Thompson (2011) and China's role in the US-India relationship (Madan 2020) are basically RAA triangles with major powers. Due to their common rival, a major-power mixed third party can support its ally to maintain its rival relationship, and thus, the focal rivalry can last longer with the help of the major-power mixed third party.

I hypothesize that a major-power third party is more likely to influence rivalry termination than a minor-power third party. Particularly, when the major-power third-party is a common ally, its mediation attempts could provide greater benefits to the focal rivalry. If a common rival is a major power, the focal rivals are more likely to prioritize the rival relationship with the major power.

Hypothesis 4: If a rivalry has more major power countries as third parties in triangular relationships, their effects on rivalry duration are stronger than non-major-power third parties.

Research Design

I test the triangular relationship and the third parties' major power status hypotheses by employing the Cox proportional hazards model with time-varying covariates. The total

number of rivalries is 371, and the temporal domain is from 1900 to 2001 (Goertz et al. 2016). The list of 371 rivalries is generated based on the start and end dates of severe and lesser rivalries in the Peace data.[6] 297 out of 371 rivalries terminated in this period. The mean rivalry duration is 21.16 years, and it ranges from 1 to 102. The longest rivalries are the Colombia-Venezuela, Bolivia-Chile, and Russia-Japan rivalries. Since there are no minimum time criteria for measuring rivalry in the Peace data, some rivalries have a shorter duration, such as less than 1 year.[7]

The first set of independent variables is triangular relationships, including rivalries and defense pacts—*RAA*, *RRA*, and *RRR*. Both severe and lesser rivalries in the Peace data are employed (Goertz et al. 2016). Defense pacts are drawn from the Alliance Treaty Obligations and Provision (ATOP) dataset (Leeds et al. 2002). ATOP conceptualizes an alliance as "a formal agreement among independent states to cooperate militarily in the face of potential or realized military conflict" (Leeds and Mattes 2007, 185). A defense pact focuses on "a promise of active military support in the event of an attack" on one or more members (Leeds and Mattes 2007, 189). To capture the three types of triangles, I create a cross-network of rivalries and defensive alliances for each year. When a potential third party has both rivalry and defense pact ties with a focal country, this third party is not counted for triangles (see Online Appendix).

In the cross-network, countries are connected via positive or negative ties. The ties in the network are undirected because if A is B's rival, then B is A's rival. For each year of a rivalry, I count the number of triangles where a rivalry is embedded.[8] For instance, the UK-USSR rivalry was embedded in four RRA triangles in 1969 with the US, France, West Germany, and Norway. Based on this information, *RRA* is coded 4 for the UK-USSR rivalry in 1969. 238 out of 371 rivalries (64.15%) had at least one third party in their rivalry history, and for those rivalries, the average numbers of triangles are 0.07 (*RAA*), 1.51 (*RRA*), and 0.51 (*RRR*).

The second set of independent variables considers the power status of third-party countries: *major-* and *minor-power RAA, RRA,* and *RRR*. All triangles are categorized based on their type and third parties' major power status. The information about major power status is drawn from the Correlates of War Project (2017). In the example of the UK-USSR rivalry, two third parties were major power (US and France), and the others were minor power (West Germany and Norway). Thus, for the UK-Russia rivalry in 1969, *RRA-major* is coded 2, and *RRA-minor* is also coded 2.

The independent and control variables vary on an annual basis. I control for net polity change, democratic dyads, and system- and state-level shocks (Bennett 1998). First, net polity change is calculated by summing up both rivals' polity change, which is drawn from the Polity IV Project (Marshall et al. 2018). Second, a country with a polity score greater than five is considered a democracy. If both countries in a rivalry are democratic, this rivalry is coded 1 for democratic dyads and 0 otherwise. For political shocks, system-level shocks are measured by a binary variable capturing the World Wars (1914-1918, 1939-1945), territorial shocks (1845-1849, 1884-1894, 1898-1903, 1956-1962), and power distribution shocks (1859-1877) (Bennett 1998; Goertz and Diehl 1995). Civil war is also measured as a binary variable (Dixon and Sarkees 2016). Considering Goertz and Diehl's (1995) finding

Table 3. Effect of Triangles on Rivalry Duration (1900-2001).

	Model 1: All	Model 2: Major-Minor Rivalry	Model 3: Minor-Minor Rivalry
Common allies (RAA)	0.015 (0.119)	−0.133 (0.213)	0.781** (0.385)
Mixed third parties (RRA)	0.042 (0.030)	0.066** (0.030)	−0.169* (0.094)
Common rivals (RRR)	0.145* (0.079)	0.242** (0.095)	−0.192 (0.200)
Controls			
Defense pact	0.257 (0.168)	0.661* (0.347)	0.145 (0.205)
Democratic dyad	0.316 (0.288)	0.789** (0.364)	−0.249 (0.458)
Net polity change	0.066** (0.027)	0.061 (0.045)	0.059** (0.029)
Issue salience (max)	−0.057*** (0.015)	−0.049* (0.026)	−0.046** (0.020)
Capability ratio	0.000 (0.000)	−0.000 (0.000)	0.000 (0.002)
Domestic conflict (log)	−0.013 (0.018)	−0.057** (0.027)	0.064** (0.030)
Civil war shock (10 years lag)	−0.166* (0.090)	−0.205 (0.156)	−0.157 (0.127)
System-level shock (10 years lag)	−0.093 (0.126)	0.192 (0.241)	−0.360* (0.188)
Observations	7852	2713	4535
Log-likelihood	−1505.882	−524.156	−643.952
Wald chi-square	43.02	42.62	31.96
Prob > chi-square	<0.0001	<0.0001	<0.001

Note: Unexponentiated coefficients reported. Standard errors in parentheses, ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

on political shocks and rivalry termination, 10-years lags are included in the model, thus, time points from the occurrence of the event to +10 years after the event ends are coded 1.

I control for defensive pacts, power ratio between rivals, domestic conflict, and issue salience.[9] Rivalry dynamics can be different in power symmetry (Vasquez 2009). The power ratio is measured using the CINC score (Singer et al. 1972). In this measure, the larger value is divided by the smaller value within a rivalry for a given year. I include the domestic conflict index of the Cross-National Time-Series Data Archive (Banks 2011). For each rivalry, the index scores are logged and summed up. Lastly, I control for the salience level of territorial issues from the Issue Correlates of War Project. The salience level ranges from 0 to 12, generated based on the tangible and intangible indicators (Frederick et al. 2017; Hensel et al. 2008). Considering the tendency of issue accumulation among rivals (Dreyer 2010), I use the maximum value of salience scores for rivalries with multiple territorial issues.

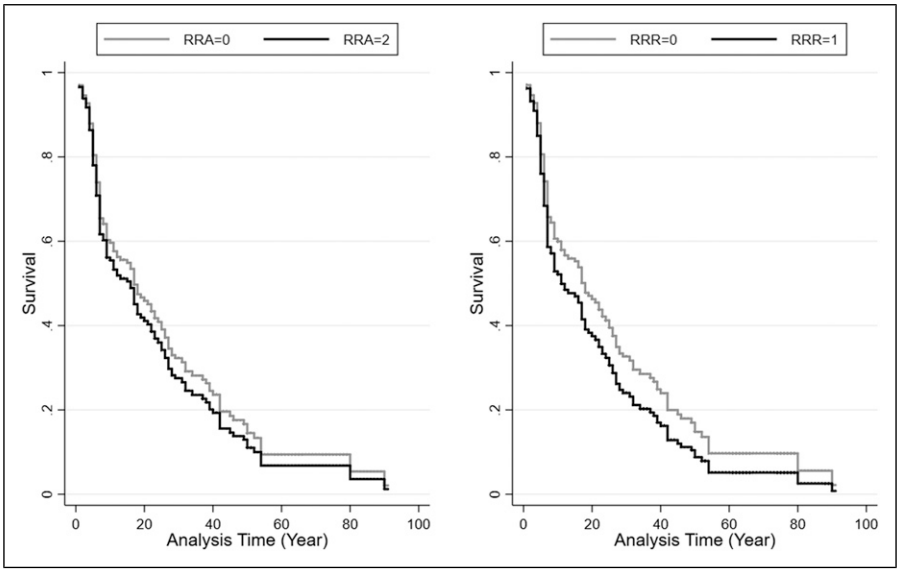


Figure 5. Survival functions from model 2, Table 3.

Empirical Findings

I expect that a rivalry with common allies (*RAA*) and common enemies (*RRR*) has a shorter duration (H1 and H2), while a rivalry with mixed third parties (*RRA*) has a longer duration (H3). I test my hypotheses using all 371 rivalries and then move to subgroups of major-minor and minor-minor rivalries (Table 3). Positive coefficients indicate that the variable leads to a shorter rivalry duration (hazard ratios ↑), while negative coefficients suggest that the variable influences a longer rivalry duration (hazard ratios ↓).

In Model 1, including all rivalries, *RAA* and *RRA* are insignificant, but *RRR* is minimally significant ($p < 0.1$). The positive coefficient of *RRR* means that common rivals increase the rivalry termination rate, leading to a shorter duration. One more common rival increases the termination rate by 15.6%. This suggests that having more common rivals helps the focal rivals end their rivalry.

When looking at subgroups of rivalries, common rivals (*RRR*) of major-minor rivalries is positive and significant (Model 2). The termination rate is 27.3% higher for major-minor rivalries with one common rival compared to those without common rivals. This supports H2 of prioritization. *RRA* is also positive and significant, which means that more mixed third parties lead to a shorter duration of major-minor rivalries. One more mixed third party increases the termination rate of major-minor rivalries by 6.8%. This does not support H3 of stable blocs, suggesting that mixed third parties do not help their connected major-minor rivalries endure.

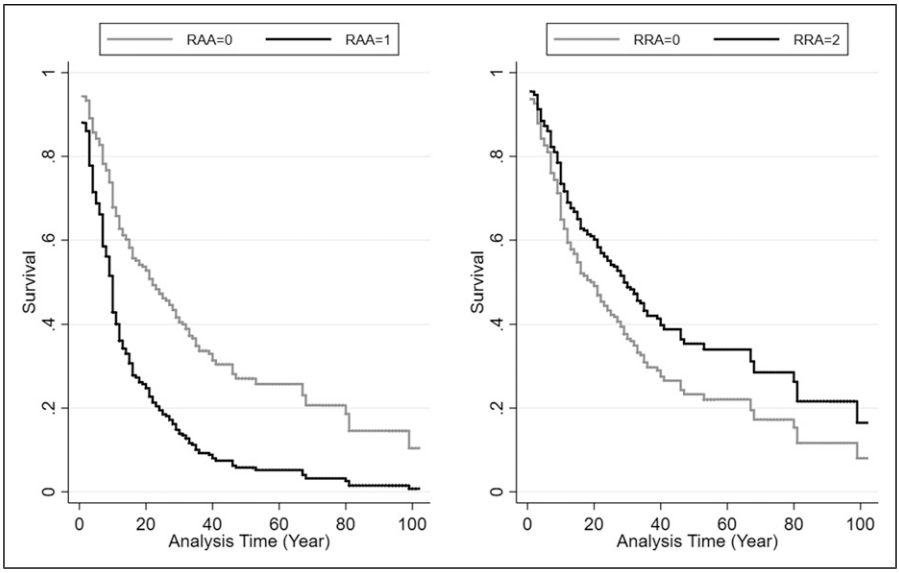


Figure 6. Survival functions for model 3, Table 3.

In Model 2, the substantive effect of *RRA* is the smallest among the variables with significant effects. Democratic dyad (0–1, 120.1% ↑) has the greatest substantive effect, followed by defense pact (0–1, 93.7% ↑). *RRR* has the third greatest effect (one more RRR, 27.3% ↑), followed by issue salience (mean to one s.d. above the mean, 19.9% ↓) and domestic conflict (mean to one s.d. above the mean, 18.3% ↓). Figure 5 shows that the survival function of major-minor rivalries with two RRAs (left, black) is below that without RRA (left, gray), and that with one RRR (right, black) is also below that without RRR (right, gray). The findings indicate that major-minor rivalries with common rivals or mixed third parties have a shorter duration compared to those without them.

Contrary to RRR's significant and substantive effects on major-minor rivalries, it is insignificant in minor-minor rivalries (Model 3). However, the coefficient of mixed (*RRA*) is negative and minimally significant ($p < 0.1$) for minor-minor rivalries, which supports H3 of stable blocs. This suggests that more mixed third parties lead to a longer duration of minor-minor rivalries. One more RRA decreases the termination rate by 15.5%. This means that mixed third parties (Zs) help their connected minor-minor rivalries last longer, and in that way, they share the cost with their ally (Y) to fight against a rival (X).

On the other hand, common allies (*RAA*) of minor-minor rivalries is positive and significant, supporting H1 of information flow. This suggests that more common allies lead to a shorter duration of minor-minor rivalries. Specifically, one more common ally increases the termination rate of minor-minor rivalries by 118.3%. In terms of the size of substantive effects, having one more common ally has the greatest effect on the duration

of minor-minor rivalries (118.3% ↑), followed by system-level shock (0–1, 30.2% ↓), domestic conflict (mean to one s.d. above the mean, 25.6% ↑), and issue salience (mean to one s.d. above the mean, 18.8% ↓). One more mixed third party (*RRA*) follows with 15.5%, and net polity change has the smallest substantive effect on minor-minor rivalry duration (mean to one s.d. above the mean, 15.5% ↑). Figure 6 (left) shows that the survival function of minor-minor rivalries with one RAA (black) is below that without RRA (gray). On the right side, the survival function of minor-minor rivalries with two RRAs (black) is above that of minor-minor rivalries with no RRA (gray). Hence, we can see that minor-minor rivalries are more influenced by third parties, particularly, and their common allies affect a shorter duration of minor-minor rivalries.[10]

Table 4. Effect of Triangles and Third-Parties' Major-Power Status on Rivalry Duration (1900-2001).

	Model 4: All	Model 5: Major-Minor Rivalry	Model 6: Minor-Minor Rivalry
RAA: Common allies			
Major power	−0.148 (0.386)	−0.215 (0.373)	0.275 (0.997)
Minor power	0.058 (0.129)	−0.056 (0.163)	0.940*** (0.352)
RRA: Mixed third parties			
Major power	0.144* (0.081)	0.028 (0.124)	−0.023 (0.149)
Minor power	0.003 0.050)	0.072 (0.072)	−0.249* (0.135)
RRR: Common rivals			
Major power	0.229* (0.125)	0.107 (0.174)	0.004 (0.336)
Minor power	0.058 (0.129)	0.370*** (0.136)	−0.288 (0.277)
Controls			
Defense pact	0.270 (0.169)	0.662* (0.346)	0.138 (0.205)
Democratic dyad	0.330 (0.287)	0.797** (0.370)	−0.253 (0.455)
Net polity change	0.065** (0.027)	0.061 (0.044)	0.059** (0.029)
Issue salience (max)	−0.055*** (0.015)	−0.051* (0.027)	−0.046** (0.020)
Capability ratio	0.000 (0.000)	−0.000 (0.000)	0.000 (0.002)
Domestic conflict (log)	−0.009 (0.018)	−0.062** (0.026)	0.067** (0.031)
Civil war shock (10 years lag)	−0.177* (0.091)	−0.202 (0.154)	−0.175 (0.128)
System-level shock (10 years lag)	−0.082 (0.127)	0.192 (0.246)	−0.352* (0.188)
Observations	7852	2713	4535
Log-likelihood	−1504.734	−523.569	−643.217
Wald chi-square	45.61	47.83	34.99
Prob > chi-square	<0.0001	<0.0001	<0.01

Note: Unexponentiated coefficients reported. Standard errors in parentheses, ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

Likewise, some basic triangle variables have significant effects on rivalry duration. Then, who are those third parties? Are they major power? Overall, the significant effects come from minor power, not from major-power third parties, which do not support H4 (Table 4). In Model 4, including all rivalries, the coefficient of major-power common rivals (*RRR*) is positive and minimally significant ($p < 0.1$), consistent with the results in Model 1. This means that rivalries with one or more major-power common rivals have a shorter rivalry duration, and one more major-power common rival increases the termination rate by 25.7%.

When looking at subgroups of rivalries, all coefficients of major-power third parties are insignificant. For major-minor rivalries, the coefficient of minor-power common rivals (*RRR*) is positive and significant, indicating that minor-power common rivals influence a shorter duration of major-minor rivalries (one more common rival, 44.8% ↑). In Model 6 of minor-minor rivalries, the coefficient of minor-power common allies (*RAA*) is positive and significant, and that of minor-power mixed (*RRA*) is negative and minimally significant ($p < 0.1$), consistent with the results in Model 3. For minor-minor rivalries, one more minor-power common ally increases the termination rate by 156.1% (shorter duration), while one more minor-power mixed third party decreases the termination rate by 22.0% (longer duration). Those results do not support H4 that major-power third parties have more influence on the duration of their connected rivalries.

Discussion

Empirical results in Models 5 and 6 show that minor-power common rivals and common allies influence major-minor and minor-minor rivalries, respectively, but not major-power third parties. I plot the survival functions using those subgroups with and without major-power third parties (Figures A2 and A3). The survival function of major-minor rivalries with one major-power and one minor-power common rivals (black, solid) is below that of no major-power and one minor-power common rival (gray, solid), but they are too close to make a significant difference. The distance between survival functions of no minor-power common rivals (black dotted and gray dotted) is also narrow. Similarly, in Figure A3, the survival function of minor-minor rivalries with one major-power and one minor-power common allies (black, solid) is below that of minor-minor rivalries with no major-power and one minor-power common ally (gray, solid), but the difference between two survival functions is insignificant.

The results in Model 2 show that mixed third parties increase the termination rate of major-minor rivalries, leading to a shorter duration. In the model, I count all mixed third parties for major-minor rivalries, but that doesn't show whose allies they are. For the focal major-minor rivalry (M-m), mixed third parties can be M's allies and m's rivals, or M's rivals and m's allies. When making this distinction, the coefficient of minor-power's allies (m's allies and M's rivals) is positive and significant (Model 1, Table A4 in Online Appendix). In other words, when a minor-power rival has more minor-power allies, its rivalry with a major power is more likely to end (shorter duration).

However, when considering the major power status of mixed third parties, none are significant (Model 2, Table A4).

In the core independent variables, defense pacts are used to capture RAA and RRA triangles. As described above, some rivalries had defense pacts at some point during their rivalry periods, and those dyads having both types of relationships are not included in triangles. However, having one or more defense pacts can influence the longevity of the rivalry, thus, the existence of defense pacts in the focal rivalry is included as a control variable in models. Considering that rival relationships can overshadow the effects of having defense pacts, I estimate models without the defense pacts variable (Tables A5 and A6). The results show that RRR is significant for major-minor rivalries, while RAA and RRA are significant for minor-minor rivalries, which are consistent with the results in Tables 3 and 4.

Conclusion

Most rivalries are connected to other rivalries and alliances in various ways, which affect the longevity of rivalries. This article introduces the overlapping immediate strategic environment by focusing on all types of triangles a focal rivalry is embedded in. By using this approach, this study shows how different types of third parties collectively influence rivalry duration, which moves beyond the traditional dyadic approach to examining rivalry duration. I also highlight how the power relationships within triangle relationships influences rivalry duration.

First, I find that the effects of triangular relationships vary across major-minor and minor-minor rivalries. Overall, minor-minor rivalries are more likely to be influenced by third parties in triangular relationships. For minor-minor rivalries such as Japan and South Korea, a common ally like the United States increases the hazard rate of rivalry termination (shorter duration) due to the third party facilitating information flows between the competitors and serving as an outside mediator. On the other hand, mixed third parties, such as China's support of North Korea in the Korean rivalry, create stable blocs and lengthen the rivalry's duration. When major powers become involved as rivalry actors, though, we see stronger effects for the presence of common enemies in triangular relationships. For major-minor rivalries, the termination rate increases if the rivalries have common enemies compared to other major-minor rivalries without common enemies.

Second, when considering major power status, the effects of major-power third parties are not strong across rivalry types. However, the number of minor-power common allies has a significant effect on the duration of minor-minor rivalries (shorter duration), and that of minor common rivals also does on the duration of major-minor rivalries. From the results, we can infer that regardless of major power status, more common allies can provide more opportunities for the rivals to end their rivalries. Also, in major-minor rivalries, even major power countries have incentives to reduce the number of rivalries with minor power countries to focus on more important rivals.

Recently, more countries have joined or are willing to join the North Atlantic Treaty Organization (NATO). The enlargement of NATO is creating more RRA triangles, which were common during the Cold War. Considering the varying effects of mixed third parties (RRA) on major-minor and minor-minor rivalries, it would be crucial to understand rivalries' strategic environments and their effects on rivalry duration. In further research, this approach can be employed to examine rivalry dynamics (escalation/de-escalation of rivalry interactions) and the formation and duration of alliances.

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Supplemental Material

Supplemental material for this article is available online.

Notes

1. The Peace data version 2.01 (Goertz et al. 2016) is employed for the percentages.
2. The numbers are drawn from the Militarized Interstate Dispute Data version 4.3 from the Correlates of War Project.
3. The ATOP data provides five types of alliances: offense, defense, neutrality, nonaggression, and consultation pacts. Among 371 rivalries from 1900 to 2001 in the Peace data, 205 rivalries (55.26%) had at least one formal alliance treaty at some point during their rivalry. For instance, the inter-Korean rivalry has had nonaggression pacts since 1991. Considering the theory in the next section and the overlapping relationships of rivalries and broadly defined alliances, this paper focuses on defense pacts instead of all types of alliances in the ATOP data.
4. This term is borrowed from Caplow (1968).

5. Positional rivalries compete over relative influence and status within a system or subsystem, while spatial rivalries compete over the control of territory (Colaresi et al. 2007).
6. When a dyad started from a severe rivalry and moved to a lesser rivalry, the transition date is not counted as termination. When a rival relationship is resumed after its negative peace relationship, two rivalry periods are counted separately.
7. For instance, some of the Warsaw Pact members had short-term rival relationships in 1968 because of the Prague Spring.
8. Since the rivalries are undirected, a triangle is defined to be three nodes with three edges, {(i, j), (j, k), (k, i)}, where i ≠ j ≠ k, such as $\sum_{(ijk)} N_{ij}^t N_{ik}^t N_{jk}^t$.
9. Overlapping relationships of rivalry and defense pacts are not considered for triangles. However, at the dyadic level, rivalries having defense pacts can be different from those without defense pacts in terms of their longevity. Thus, I control for defense pacts in models.
10. The empirical results show that the characteristics of major-minor rivalries are different from those of minor-minor rivalries. Only issue salience is consistently negative and significant for both sub-groups. Domestic conflict is negative in major-minor rivalries, while it is positive in minor-minor rivalries. Defense pacts and democratic dyads are significant in major-minor rivalries, whereas net polity changes and system-level shocks are significant in minor-minor rivalries.

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